



SCHOOL OF COMPUTER SCIENCES
UNIVERSITI SAINS MALAYSIA

COURSE-CODE - Course Title

Semester X, Academic Session YYYY/YYYY

Assignment Title

Report Topic Title

Lecturer: Lecturer Name

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The sections listed in this template are only a reference structure. Adjust, rename, add, or remove sections according to the assignment brief and lecturer requirements. Keep References as the final item in the Table of Contents unless the brief states otherwise.

1.0 Introduction

Briefly describe the assignment, the problem being solved, and the overall approach. Cite external sources using IEEE-style numbered citations such as [1]. A good introduction usually answers:

- What is the assignment about?
- What problem or scenario is being addressed?
- What are the main methods, algorithms, or tools used?

2.0 Objectives

State the goals of the assignment clearly. For example:

1. Implement the required algorithm or system component.
2. Analyze correctness, performance, or usability.
3. Evaluate the output using suitable test cases or examples.

3.0 Background

Summarize the key concepts, theories, data structures, algorithms, or technologies needed to understand the work. Keep this section focused on concepts directly used in the assignment.

4.0 Methodology

Explain the approach step by step. For algorithmic assignments, include the high-level design, assumptions, input/output format, and complexity considerations.

4.1 System or Algorithm Design

Describe the design using text, diagrams, tables, or pseudocode.

Table 1: Example table format

Component	Purpose	Notes
Input parser	Reads input data	Validate edge cases
Solver	Runs main algorithm	Analyze complexity
Output formatter	Displays result	Keep format consistent

Algorithm 1 Process Input Data

```

1: procedure SOLVE(inputData)
2:   result  $\leftarrow$  [ ]
3:   for all item  $\in$  inputData do
4:     append PROCESS(item) to result
5:   end for
6:   return result
7: end procedure

```

4.2 Pseudocode

5.0 Implementation

Describe the implementation details that matter for understanding the work. Avoid pasting every line of source code unless the assignment requires it. Focus on important functions, modules, decisions, and edge cases.

5.1 Important Code Fragment

```

1 def process(item):
2     if item is None:
3         return "invalid"
4     return str(item).strip()

```

Listing 1: Replace with a relevant code fragment

6.0 Testing and Results

Explain how the solution was tested. Include meaningful cases such as normal inputs, boundary cases, invalid inputs, and larger examples.

Table 2: Example test cases

Test case	Input summary	Expected result	Actual result
TC1	Basic valid input	Correct output	Correct output
TC2	Empty input	Graceful handling	Graceful handling
TC3	Large input	Completes successfully	Completes successfully

7.0 Discussion

Discuss the results. Mention what worked well, limitations, trade-offs, and any assumptions that affected the outcome.

8.0 Conclusion

Summarize the work completed, the main findings, and what could be improved in future work.

9.0 References

- [1] T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*, 3rd ed. Cambridge, MA, USA: MIT Press, 2009.
- [2] Overleaf, "LaTeX documentation." [Online]. Available: <https://www.overleaf.com/learn>. [Accessed: Jun. 12, 2026].